14.2: Probability laws govern Mendelian inheritance

Mendel's laws of segregation and independent assortment reflect the same rules of probability that apply to tossing coins, rolling dice, and drawing cards from a deck. The probability scale ranges from 0 to 1. An event that is certain to occur has a probability of 1, while an event that is certain *not* to occur has a probability of 0. With a coin that has heads on both sides, the probability of tossing heads is 1, and the probability of tossing tails is 0. With a normal coin, the chance of tossing heads is 12, and the chance of tossing tails is 12. The probability of drawing the ace of spades from a 52-card deck is 152. The probabilities of all possible outcomes for an event must add up to 1. With a deck of cards, the chance of picking a card other than the ace of spades is 5152.

Tossing a coin illustrates an important lesson about probability. For every toss, the probability of heads is 12. The outcome of any particular toss is unaffected by what has happened on previous trials. We refer to phenomena such as coin tosses as independent events. Each toss of a coin, whether done sequentially with one coin or simultaneously with many, is independent of every other toss. And like two separate coin tosses, the alleles of one gene segregate into gametes independently of another gene's alleles (the law of independent assortment). We'll now look at two basic rules of probability that help us predict the outcome of the fusion of such gametes in simple monohybrid crosses and more complicated crosses as well.

The Multiplication and Addition Rules Applied to Monohybrid Crosses

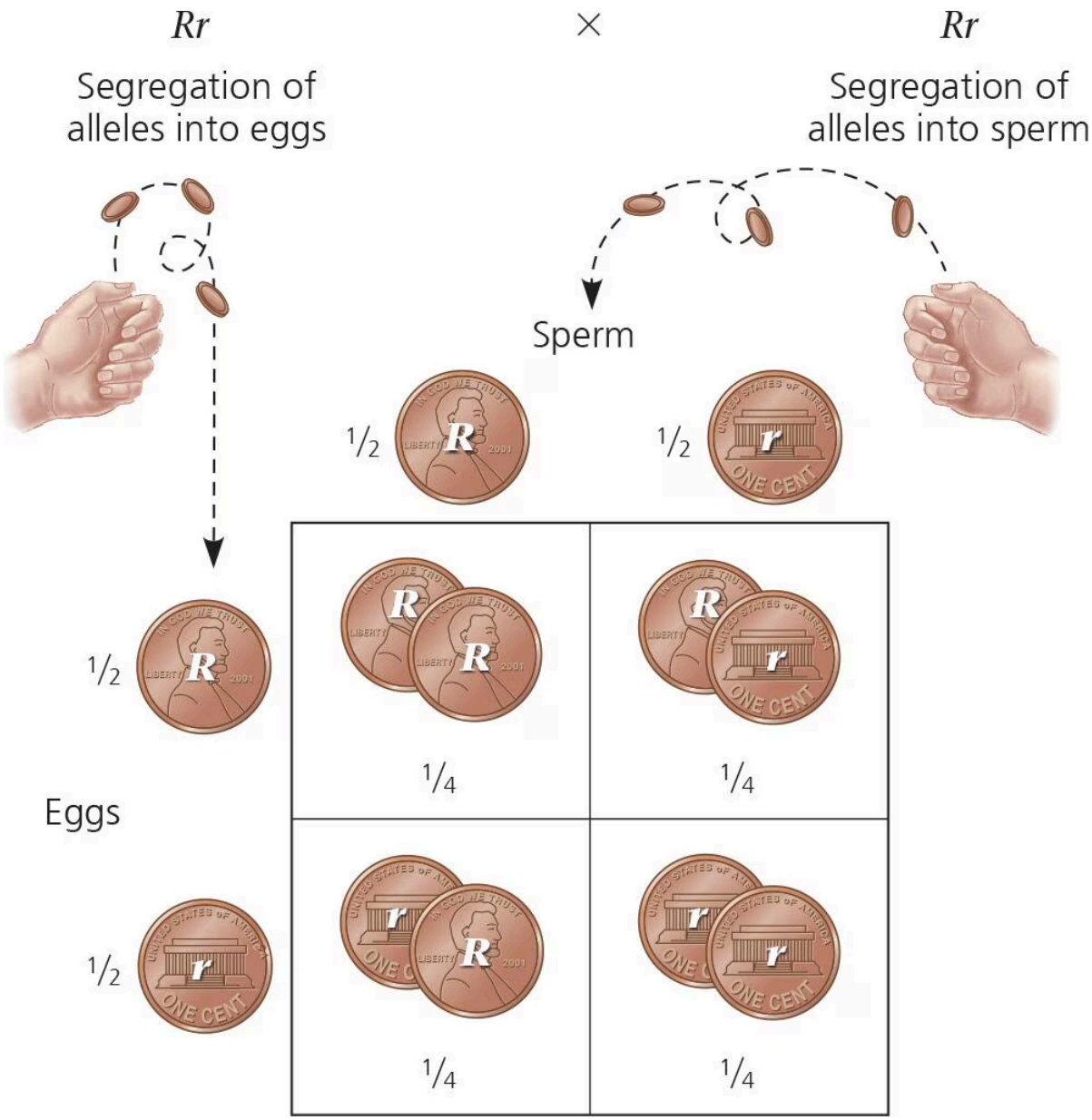
How do we determine the probability that two or more independent events will occur together in some specific combination? For example, what is the chance that two coins tossed simultaneously will both land heads up? The **multiplication rule** states that to determine the probability of one event *and* the other occurring, we multiply the probability of one event (one coin coming up heads) by the probability of the other event (the other coin coming up heads). By the multiplication rule, then, the probability that both coins will land heads up is 12×12=14.

We can apply the same reasoning to an F1 monohybrid cross. With seed shape in pea plants as the heritable character, the genotype of F1 plants is Rr. Segregation in a heterozygous plant is like flipping a coin in terms of calculating the probability of each outcome: Each egg produced has a 12 chance of carrying the dominant allele (R) and a 12 chance of carrying the recessive allele (r). The same odds apply to each sperm cell produced. For a particular F2 plant to have wrinkled seeds, the recessive trait, both the egg and the sperm

that come together must carry the r allele. The probability that an r allele will be present in the egg *and* in the sperm at fertilization is found by multiplying 12 (the probability that the egg will have an r)×12 (the probability that the sperm will have an r). Thus, the multiplication rule tells us that the probability of an F2 plant having wrinkled seeds (rr) is 14**(**Figure 14.9**)**. Likewise, the probability of an F2 plant carrying both dominant alleles for seed shape (RR) is 14.

Figure 14.9

Segregation of alleles and fertilization as chance events.



Gametes	one-half uppercase R (heads)	one-half lowercase r (tails)
one-half uppercase R (heads)	RR both uppercase, both coins are heads	Uppercase R lowercase
	Probability = one-fourth	One heads, one tails
		Probability = one-fourth
one-half lowercase r (tails)	Uppercase R lowercase r	rr both lowercase, both coins are tails
	One heads, one tails	Probability = one-fourth
	Probability = one-fourth	

Alleles from upper case R lowercase r parent segregate into eggs, while alleles from another uppercase R lowercase r parent segregate into sperm. Half of the gametes from each parent will be uppercase R and half will be lowercase r, just as there is a one-half chance of flipping heads and a one-half chance of flipping tails when flipping a coin. The gametes (eggs and sperm) are superimposed on coins (heads or tails) in a Punnett square with two rows and two columns. The eggs (left side) are one-half uppercase R (heads) and one-half lowercase r (tails). The sperm (top) are one-half uppercase R (heads) and one-half lowercase r (tails). A Punnett square is represented by the following table.

When a heterozygote (Rr) forms gametes, whether a particular gamete ends up with an R or an r is like the toss of a coin. We can determine the probability for any genotype among the offspring of two heterozygotes by multiplying together the individual probabilities of an egg and sperm having a particular allele (R or r in this example).

To figure out the probability that an F2 plant from a monohybrid cross will be heterozygous rather than homozygous, we need to invoke a second rule. Notice in Figure 14.9 that the dominant allele can come from the egg and the recessive allele from the sperm, or vice versa. That is, F1 gametes can combine to produce Rr offspring in two *mutually exclusive* ways: For any particular heterozygous F2 plant, the dominant allele can come from the egg *or* the sperm, but not from both. According to the **addition rule**, the probability that any one of two or more mutually exclusive events (one event *or* the other) will occur is calculated by adding their individual probabilities. As we have just seen, the multiplication rule gives us the individual probabilities that we will now add together. The probability for one possible way of obtaining an F2 heterozygote—the dominant allele from the egg and the recessive allele from the sperm—is 14. The probability for the other possible way—the recessive allele from the egg and the dominant allele from the sperm—is also 14 (see Figure 14.9). Using the rule of addition, then, we can calculate the probability of an F2 heterozygote as 14+14=12. (Note that use of the word *and* in the statement is a clue that you should use the multiplication rule, while *or* suggests use of the addition rule.)

Solving Complex Genetics Problems with the Rules of Probability

We can also apply the rules of probability to predict the outcome of crosses involving multiple characters. Recall that each allelic pair (gene) segregates independently during gamete formation (the law of independent assortment). Thus, a dihybrid or other multicharacter cross is equivalent to two or more independent monohybrid crosses occurring simultaneously. By applying what we have learned about monohybrid crosses, we can determine the probability of specific genotypes occurring in the F2 generation without having to construct unwieldy Punnett squares.

Consider the dihybrid cross between YyRr heterozygotes shown in Figure 14.8. We will focus first on the seed-color character. For a monohybrid cross of Yy plants, we can use a simple Punnett square to determine that the probabilities of the offspring genotypes are 14 for YY, 12 for Yy, and 14 for yy. We can draw a second Punnett square to determine that the same probabilities apply to the offspring genotypes for seed shape: 14 RR,12Rr, and 14rr. Knowing these probabilities, we can simply use the multiplication rule to determine the probability of each of the genotypes in the F2 generation. To give two examples, the calculations for finding the probabilities of two of the possible F2 genotypes (YYRR and YyRR) are shown below:

$$\text{Probability of } YYRR = \tfrac{1}{4}\,(\text{probability of } YY) \times \tfrac{1}{4}\,(RR) = \tfrac{1}{16}$$

$$\text{Probability of } YyRR = \tfrac{1}{2}\,(Yy) \times \tfrac{1}{4}\,(RR) = \tfrac{1}{8}$$

The YYRR genotype corresponds to the upper left box in the larger Punnett square in Figure 14.8 (one box=116). Looking closely at the larger Punnett square in Figure 14.8, you will see that 2 of the 16 boxes (18) correspond to the YyRR genotype.

Now let's see how we can combine the multiplication and addition rules to solve even more complex problems in Mendelian genetics. Imagine a cross of two pea varieties in which we track the inheritance of three characters. Let's cross a trihybrid with purple flowers and yellow round seeds (heterozygous for all three genes) with a plant with purple flowers and green wrinkled seeds (heterozygous for flower color but homozygous recessive for the other two characters). Using Mendelian symbols, our cross is PpYyRr×Ppyyrr. What fraction of offspring from this cross are predicted to exhibit the recessive phenotypes for *at least two* of the three characters?

To answer this question, we can start by listing all genotypes we could get that fulfill this condition: ppyyRr, ppYyrr, Ppyyrr*,*PPyyrr, and ppyyrr. (Because the condition is *at least two* recessive traits, it includes the last genotype, which shows all three recessive traits.) Next, we calculate the probability for each of these genotypes resulting from our PpYyRr×Ppyyrr cross by multiplying together the individual probabilities for the allele pairs, just as we did in our dihybrid example. Note that in a cross involving heterozygous and

homozygous allele pairs (for example, Yy×yy), the probability of heterozygous (Yy) offspring is 12 and the probability of homozygous (in this case, yy) offspring is 12. Finally, we use the addition rule to add the probabilities for all the different genotypes that fulfill the condition of at least two recessive traits resulting from our PpYyRr×Ppyyrr cross, as shown below:

$$
\begin{array}{ll}
ppyyRr & \tfrac{1}{4}\,(\text{probability of } pp) \times \tfrac{1}{2}\,(yy) \times \tfrac{1}{2}\,(Rr) = \tfrac{1}{16} \\
ppYyrr & \tfrac{1}{4}\,(pp) \times \tfrac{1}{2}\,(Yy) \times \tfrac{1}{2}\,(rr) = \tfrac{1}{16} \\
Ppyyrr & \tfrac{1}{2}\,(Pp) \times \tfrac{1}{2}\,(yy) \times \tfrac{1}{2}\,(rr) = \tfrac{2}{16} \\
PPyyrr & \tfrac{1}{4}\,(PP) \times \tfrac{1}{2}\,(yy) \times \tfrac{1}{2}\,(rr) = \tfrac{1}{16} \\
ppyyrr & \tfrac{1}{4}\,(pp) \times \tfrac{1}{2}\,(yy) \times \tfrac{1}{2}\,(rr) = \tfrac{1}{16} \\
\hline
\text{Chance of } \textit{at least two} \text{ recessive traits} & = \tfrac{6}{16} = \tfrac{3}{8}
\end{array}
$$

The calculations are as follows.

probability lower p p y y upper R lower r = one fourth, probability of lower p p, times one half, lower y y, times one half uppercase R lower r equals to one-sixteenth.

probability lower p p upper Y lowercase y r r equals to one fourth, lower p p, times one half, uppercase Y lowercase y, times one half, lowercase r r equals to one-sixteenth.

probability uppercase P lowercase p y y r r equals to one half, uppercase P lowercase p, times one half lowercase y y, times one half lowercase r r equals to two sixteenths.

probability uppercase P P lowercase y y r r equals to one-fourth, upper P P, times one half, lowercase y y, times one half, lowercase r r, equals to one-sixteenth.

probability lowercase p p y y r r equals to one-fourth, lowercase p p, times one-half, lowercase y y, times one half, lowercase r r equals to one-sixteenth.

In time, you'll be able to solve genetics problems faster by using the rules of probability than by filling in Punnett squares.

We cannot predict with certainty the exact numbers of progeny of different genotypes resulting from a genetic cross. But the rules of probability give us the *likelihood* of various outcomes. Usually, the larger the sample size, the closer the results will conform to our predictions. Mendel understood this statistical feature of inheritance and had a keen sense of the rules of chance. It was for this reason that he set up his experiments so as to generate, and then count, large numbers of offspring from his crosses.